PGSMC: Prototype-Guided Supervised Momentum Contrastive Learning for Hyperspectral Cross-Domain Few-Shot Classification

Lingyu Kong, Xudong Sun, *Member, IEEE*, Zifei Zhao, Jiahua Zhang, and Xiaodi Shang, *Member, IEEE*

***Abstract*—Contrastive learning (CL) has recently demonstrated great potential in hyperspectral image (HSI) few-shot classification. However, conventional methods mainly emphasize instance-level similarity while neglecting class structure, often leading to the separation of intraclass samples. Moreover, the lack of explicit class prototype modeling often leads to ambiguous decision boundaries. To address these issues, this article proposes a prototype-guided supervised momentum contrastive (PGSMC) learning for hyperspectral cross-domain few-shot classification. PGSMC first designs an asymmetric augmentation module (AAM) to enhance sample diversity by applying distinct augmentation strategies and encoders to the query and key views. Subsequently, a momentum queue (MQ) is employed to store historical key features and their corresponding labels. This mechanism enables smooth updates of class-level momentum prototypes (MPs) and mitigates the limitations imposed by mini-batch training. Finally, an MP contrastive loss is formulated to guide the model toward class-level feature representations, thereby promoting more discriminative decision boundaries. Overall, PGSMC enables query samples to contrast with more representative MPs, enhancing interclass separability and promoting well-defined class boundaries. Extensive experiments on five HSI datasets demonstrate that PGSMC significantly outperforms existing few-shot learning (FSL) methods.**

***Index Terms*—Contrastive learning (CL), cross-domain learning, few-shot classification, hyperspectral imagery, prototype CL.**

I. Introduction

HYPERSPECTRAL images (HSIs) are a special type of image data, typically consisting of tens to hundreds of contiguous spectral bands. Compared to conventional RGB images, they contain much richer spectral information. HSI classification (HSIC) is performed by assigning class labels to pixels based on their spectral characteristics, enabling precise land cover mapping. In recent years, HSIC has attracted growing interest and has been widely applied in urban planning [1], precision agriculture [2], food quality control [3], and environmental monitoring [4].

In recent decades, traditional machine learning methods have been widely used in various remote sensing tasks due to their adaptability to high-dimensional data and strong generalization ability [5]. Representative approaches include support vector machines (SVMs) [6], random forests (RFs) [7], k-nearest neighbors (KNNs) [8], and naive Bayes classifiers [9]. These methods classify land cover types by modeling spectral features of hyperspectral data. However, they typically rely on handcrafted features and shallow representations, making it difficult to capture deep spatial–spectral information [10].

Deep learning has gained wide attention for its powerful feature learning capabilities [11]. Chen et al. [12] proposed a 3D-CNN feature extractor with regularization to improve HSIC performance. Roy et al. [13] introduced hybrid spectral CNN (HybridSN), which combines 3D-CNN and 2D-CNN to capture more abstract spatial features. Liu et al. [14] proposed the center attention network (CAN) based on the scaled dot-product center attention (SDPCA) module, which achieved improved classification performance across multiple datasets. To further address the issues of scale variation and unequal contributions of class features, Liu et al. [15] proposed the multiarea target attention (MATA) model. Hong et al. [16] integrated CNN and graph convolutional network (GCN) to extract complementary features, enhancing both accuracy and scalability in HSIC. Despite these advances, deep learning models heavily rely on large amounts of labeled data, while manual annotation of HSI remains costly [17], [18].

Recently, few-shot learning (FSL) has attracted attention for its ability to learn and generalize from very limited samples [19]. FSL is essentially a metalearning approach that acquires transferable knowledge from diverse tasks to adapt quickly to new ones. Typically, it operates under the N-way K-shot setting, using K labeled samples per class to train an N-class classifier [20]. Metric learning is often applied to distinguish classes in an embedding space. To address the performance drop caused by limited labeled data, FSL has been introduced into HSIC. Current approaches are mainly divided into intradomain learning and cross-domain learning [21].

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Lingyu Kong, Xudong Sun, Zifei Zhao, and Xiaodi Shang are with the College of Computer Science and Technology, Qingdao University, Qingdao 266071, China (e-mail: konglingyu@qdu.edu.cn; sunxudong@qdu.edu.cn; zhaozifei@qdu.edu.cn; shangxd@qdu.edu.cn).

Jiahua Zhang is with the College of Computer Science and Technology, Qingdao University, Qingdao 266071, China, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhangjh@radi.ac.cn).

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Intradomain FSL methods focus on fully exploiting information within the same domain to improve classification performance. For example, Xue et al. [22] proposed a lightweight spectral–spatial Siamese network (S3Net), which enhances HSIC under few-shot settings through pairwise sample augmentation and joint loss design. Wang et al. [23] introduced a weakly supervised method based on intradomain sample expansion, combining continuous side-window filtering and a dual-branch pseudo-labeling strategy to improve accuracy. Zeng et al. [24] proposed a multistage relation network with dual metric (DM-MRN) for few-shot HSI, which adaptively integrates image-to-class (I2C) and image-to-image (I2I) relations. Liu et al. [25] proposed the semantic segmentation of HSI (SegHSI) model, which integrates clustering attention with teacher–student consistency learning to effectively improve the utilization of limited samples.

Intradomain methods usually assume that prior and target samples come from the same domain, captured by the same sensor under similar conditions. However, this assumption is often unrealistic in practice [26]. To improve generalization in unseen or mismatched domains, cross-domain learning has become a key research focus. Cross-domain FSL methods aim to leverage abundant source domain data and knowledge to address the scarcity of target samples [27]. These methods can be roughly categorized into three groups: transfer learning (TL), domain adaptation (DA), and contrastive learning (CL).

TL-based methods, such as deep FSL (DFSL) [28], regularized fine-tuning FSL (MRFSL) [29], and heterogeneous FSL (HFSL) [30], first pretrain models on the source domain and then fine-tune them with limited target samples to reduce domain discrepancy. DA-based methods, including deep cross-domain FSL (DCFSL) [31], adaptive domain-adversarial FSL (ADAFSL) [32], dual-branch DA-FSL [33], and multilevel prototype alignment (MLPA) [34], aim to align feature distributions between domains step by step to mitigate domain shift. Although TL and DA methods have shown promise in cross-domain FSL, they face challenges. TL methods rely on a few labeled target samples for fine-tuning, which may lead to overfitting under extremely limited data [35]. DA methods focus on global distribution alignment but often lack explicit modeling of class-level semantics, leading to class confusion [36].

In contrast, CL-based methods learn more discriminative embedded representations by constructing contrastive sample pairs. They show stronger robustness and generalization in few-shot scenarios. For example, Zhang et al. [37] proposed a lightweight CL-based FSL model that achieves high classification accuracy with limited labels. Li et al. [38] designed a self-supervised contrastive framework to extract class-invariant features from novel samples and improve discrimination through joint FSL training. Ye et al. [39] proposed the graph convolution contrastive FSL (GCC-FSL) method, which leverages graph convolution-based CL for cross-domain alignment and significantly improves few-shot performance in HSIC. To further improve class separability and target discrimination, Xi et al. [40] incorporated a hybrid supervised–unsupervised contrastive loss into a cascaded CNN-Transformer model, effectively reducing overfitting and class confusion in few-shot training.

Traditional CL has shown success in FSL by extracting discriminative features without relying on large labeled datasets. However, most methods rely on instance-level comparisons using positive and negative pairs, which present two main limitations in practice. First, they lack explicit class-level constraints, leading to poor intraclass compactness and severe interclass confusion. Second, in small-batch or low-resource settings, the limited number of sample pairs significantly reduces the effectiveness and robustness of CL.

To address these issues, this article proposes prototype-guided supervised momentum contrastive (PGSMC) learning for hyperspectral cross-domain few-shot classification. First, an asymmetric augmentation module (AAM) is integrated into PGSMC, where distinct augmentation strategies and encoders are applied to query and key views to enhance sample diversity. Second, a label-aware momentum queue (MQ) is designed to accumulate historical representations, with a momentum-updated key encoder used to ensure temporal stability. Finally, a prototype-guided contrastive loss is introduced, where momentum prototypes (MPs) serve as contrastive targets to explicitly enhance intraclass consistency and interclass separability.

The main contributions of this article are as follows.

1) An asymmetric augmentation strategy is employed, where queries undergo weak augmentation and keys are strongly augmented. The strong augmentation is applied through a momentum-updated key encoder, with the augmentation intensity controlled by the spatial–spectral masking ratio and the level of noise introduced. This approach enhances feature diversity and robustness without compromising structural integrity.
2) To support long-term memory during training, an MQ is introduced. This queue stores historical key features and their corresponding labels, which are used to compute class-level MPs. The mechanism is designed to increase the number of contrastive sample pairs and facilitate smoother updates to the prototype representations.
3) An MP contrastive loss is introduced, where class prototypes guide feature learning at the class level. This design addresses the lack of semantic structure typically observed in traditional instance-level CL, thereby improving both intraclass compactness and interclass separability.

The rest of this article is organized as follows. Section II reviews related work. Section III describes the PGSMC framework and core modules. Section IV presents experimental results and analysis. Section V discusses and analyzes key issues of the study. Section VI concludes the article.

II. Related Work

A. FSL

FSL aims to train models with only a small number of labeled samples, enabling rapid learning of new concepts and accurate prediction on unseen tasks. This is particularly important in fields like HSI classification and medical image

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analysis, where collecting large-scale labeled data is costly and difficult. Recent FSL research mainly focuses on two approaches: metalearning and metric learning.

Metalearning does not aim to learn a direct input–output mapping, but rather a general learning strategy or algorithm. By training on many sampled few-shot tasks, the model learns to adapt quickly to new tasks with limited data [41]. Recent studies have shown the potential of metalearning in FSL. Liu et al. [42] proposed pruning-net, a metanetwork that generates pruned weights for efficient channel selection, improving performance under limited data. Sun et al. [43] introduced meta-TL (MTL), which adapts deep models by learning scaling and shifting functions for network weights. Lee et al. [44] used a trainable linear classifier as a base learner to learn more generalizable features. These methods promote the application of metalearning in few-shot image recognition and natural language processing, offering new solutions to data scarcity.

Metric learning is intended to learn an effective embedding function that maps input samples into a low-dimensional feature space, where similar samples are placed closer together while dissimilar ones are pushed farther apart, thereby establishing a reliable and discriminative distance metric. During classification, this approach is typically performed by computing the distances between query samples and class centers [45]. Metric learning has achieved notable progress in FSL. Snell et al. [46] proposed prototypical networks, which learn a metric space where classification is based on distances to class prototypes. Sung et al. [47] extended this idea with relation networks, which learn relation scores between sample pairs using a deep distance module. Li et al. [48] further extended the triplet network into a deep k-tuplet network, where more complex sample relationships are modeled to enhance feature embedding.

B. CL

CL is a self-supervised approach that learns discriminative representations by comparing similar and dissimilar sample pairs. It aims to pull positive pairs closer and push negative pairs apart in the feature space, enabling effective feature extraction from large-scale unlabeled data.

In recent years, CL has achieved remarkable success in computer vision, natural language processing, and multimodal learning. Early methods like Siamese networks [49] use weight-sharing architectures to compute similarity between inputs, pulling positives together and pushing negatives apart. Triplet networks [50] extend this idea using anchor, positive, and negative samples, encouraging the model to keep positive samples closer to the anchor than negatives by minimizing triplet loss. A simple framework for CL of visual representations (SimCLR) [51] uses heavy data augmentation to generate positive pairs from the same image and treats other samples in the batch as negatives, enabling strong representation learning without labels. Supervised CL (SCL) [52] further improves class separation by using labels to define all same-class samples as positives and others as negatives. Moreover, to overcome the computational and memory overhead of large-batch training, the unsupervised momentum-based contrastive (MoCo) learning method [53] allows negative samples to be updated dynamically during training, thereby significantly expanding the pool of negatives. Contrastive language-image pretraining (CLIP) [54] and a large-scale image and noisy-text embedding (ALIGN) [55] extend CL to joint image–text representation, significantly improving tasks like cross-modal retrieval and image captioning, and showcasing CL's potential in multimodal learning.

III. Proposed PGSMC

As shown in Fig. 1, PGSMC consists of four key modules: the embedding module (EM), cross-domain learning module (CDLM), AAM, and MP contrastive module (MPCM). These modules are designed to extract discriminative HSI features, perform FSL while reducing domain shift, enhance view diversity, and construct MPs for CL.

A. Overall Framework

The overall framework of PGSMC is shown in Fig. 1. It consists of four components.

1) *EM:* Includes two mapping layers and a residual feature extractor. Mapping layers are first applied to both source and target domains to unify feature dimensions. Then, a shared residual network extracts spectral–spatial features from both domains.
2) *CDLM:* Based on features from EM, FSL is performed by computing distances between support and query samples in both domains. CORAL [56] is then applied to both domains to reduce domain shifts and learn transferable knowledge.
3) *AAM:* Introduces a momentum-updated key encoder alongside the query encoder, using asymmetric augmentation to increase sample diversity. Two views are generated from the same input and fed into different encoders with distinct augmentations to enhance view diversity.
4) *MPCM:* Uses an MQ to store features from key views and dynamically builds class prototypes. Serving as contrastive targets in the embedding space, these prototypes guide the model to learn more discriminative class-level features, thereby enhancing intraclass compactness and interclass separability.

B. EM

The EM consists of two mapping layers and a residual feature extractor. Each hyperspectral patch is first passed through the mapping layers for dimensional alignment, then fed into the residual extractor to obtain spectral–spatial features. Specifically, both source and target domains use 1×1 2-D convolution (Conv2D) with ch input channels and 100 output channels to map $7 \times 7 \times$ ch inputs to a unified spectral dimension.

The output is then processed by a residual extractor. As shown in Fig. 2, the network includes a spectral branch and a spatial branch, composed of 3-D convolutions (Conv3D), batch normalization (BN), ReLU, fully connected (FC), dropout,

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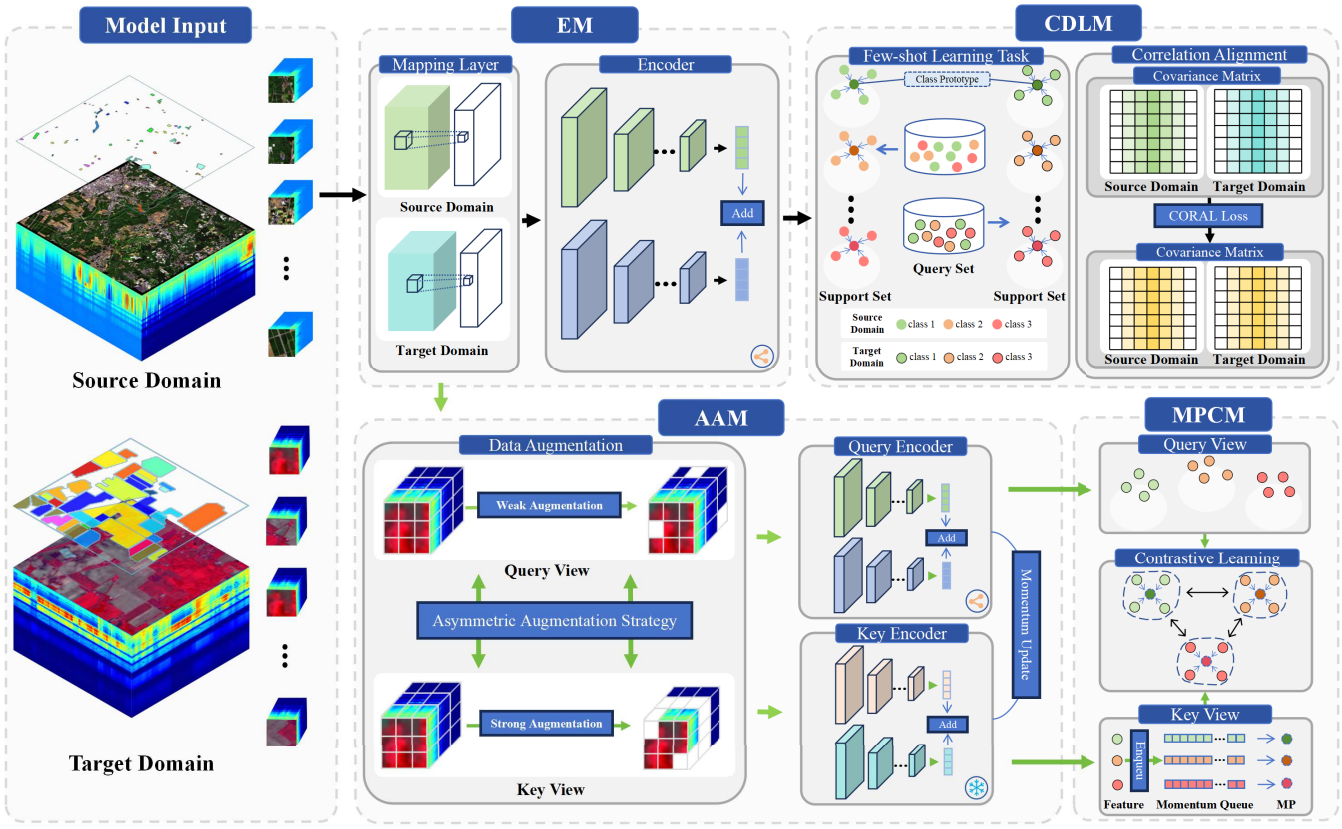


Fig. 1. Overall framework of PGSMC. The black arrows indicate the paths executed in both the source and target domains, while the green arrows indicate the paths executed only in the target domain.

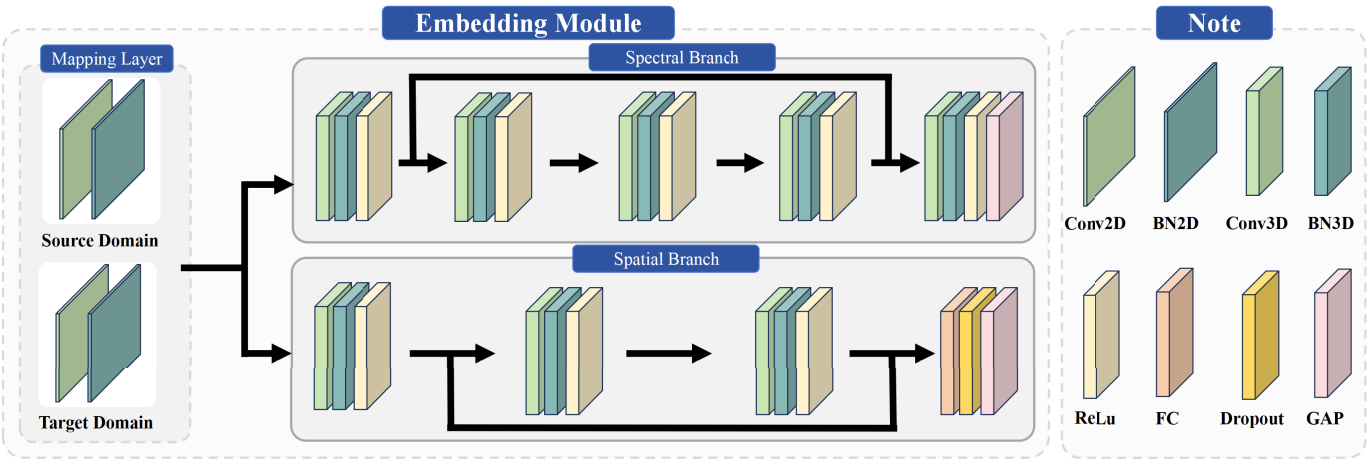


Fig. 2. Embedding module.

and global average pooling (GAP). Detailed settings are listed in Tables I and II. The spectral branch captures spectral features for distinguishing land cover types, while the spatial branch focuses on both global and local spatial patterns. Their outputs are fused to enhance feature discrimination and improve classification accuracy.

C. CDLM

The cross-domain learning module is first applied to perform FSL separately on the source and target domains, constructing a discriminative feature space with clear interclass distances. It then applies correlation alignment (CORAL) to reduce covariance shift between domains. This module includes two parts: the FSL task and CORAL.

1) FSL Task: Since the FSL process is similar for both domains, we take the source domain as an example. For each class, K labeled samples are randomly selected to form the support set $\mathbf{S}_s = \{(\mathbf{x}_i, y_i)\}_{i=1}^{C\times K}$, where C is the number of classes, $\mathbf{x}_i$ is the ith sample, and y_i is the corresponding label. Then, N additional samples per class are selected to form the

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TABLE I
DETAILS OF THE SPECTRAL BRANCH

Name	Kernel Size	Output Size
Input	-	(7 × 7 × 100)
Conv-BN-ReLU-1	(1 × 1 × 5)	(7 × 7 × 50, 32)
Conv-BN-ReLU-2	(1 × 1 × 5)	(7 × 7 × 50, 32)
Conv-BN-ReLU-3	(1 × 1 × 5)	(7 × 7 × 50, 32)
Add	-	(7 × 7 × 50, 32)
Conv-BN-ReLU-4	(1 × 1 × 5)	(7 × 7 × 50, 32)
Conv-BN-ReLU-5	(1 × 1 × 50)	(7 × 7 × 1, 128)
GAP	-	(1 × 128)

TABLE II
DETAILS OF THE SPATIAL BRANCH

Name	Kernel Size	Output Size
Input	-	(7 × 7 × 100)
Conv-BN-ReLU-1	(1 × 1 × 100)	(7 × 7 × 1, 32)
Conv-BN-ReLU-2	(3 × 3 × 1)	(7 × 7 × 1, 32)
Conv-BN-ReLU-3	(3 × 3 × 1)	(7 × 7 × 1, 32)
Add	-	(7 × 7 × 1, 32)
FC-Dropout-GAP	-	(1 × 128)

query set $\mathbf{Q}_s = \{(\mathbf{x}_j, y_j)\}_{j=1}^{C\times N}$. Both $\mathbf{S}_s = \{(\mathbf{x}_i, y_i)\}_{i=1}^{C\times K}$ and $\mathbf{Q}_s = \{(\mathbf{x}_j, y_j)\}_{j=1}^{C\times N}$ are fed into the EM for dimensionality reduction and feature extraction. Finally, sample similarity is calculated using Euclidean distance to complete the FSL tasks in both domains. It is worth noting that, due to the limited number of labeled samples in the target domain, Gaussian noise with a standard deviation of 0.04 is added when splitting the support and query sets to ensure sufficient sample size. In the source domain, the class distribution of query samples $\mathbf{x}_j$ in the query set $\mathbf{Q}_s$ is as follows:

$$P\left(y_j = u \middle| \mathbf{x}_j \in \mathbf{Q}_s\right) = \frac{\exp\left(-d\left(f_\varphi\left(\mathbf{x}_j\right), \mathbf{p}_u\right)\right)}{\sum_{c=1}^{C}\exp\left(-d\left(f_\varphi\left(\mathbf{x}_j\right), \mathbf{p}_c\right)\right)}. \quad (1)$$

In (1), d denotes the Euclidean distance, and $\mathbf{p}_c$ is the prototype of class c in the support set. f_φ represents the feature extractor with learnable parameters. Based on the extracted feature $f_\varphi(\mathbf{x})$ from a query sample and its ground-truth label u, the FSL loss in the source domain is defined as follows:

$$\mathcal{L}_{\text{fsl}}^{s} = \frac{1}{B}\left[-\sum_{(\mathbf{x},y)\in\mathbf{Q}_s}\log p_\varphi\left(y = u | f_\varphi\left(\mathbf{x}\right)\right)\right]. \quad (2)$$

Here, p_φ denotes the probability that the embedded feature belongs to class k, and B denotes the number of samples in a batch. Similarly, for each target batch, the FSL loss is computed as follows:

$$\mathcal{L}_{\text{fsl}}^{t} = \frac{1}{B}\left[-\sum_{(\mathbf{x},y)\in\mathbf{Q}_t}\log p_\varphi\left(y = u | f_\varphi\left(\mathbf{x}\right)\right)\right] \quad (3)$$

where $\mathbf{S}_t$ and $\mathbf{Q}_t$ represent the support and query sets in the target domain. This method allows the model to learn a feature space where related samples are close and unrelated samples are distant. The final FSL loss is defined as follows:

$$\mathcal{L}_{\text{fsl}} = \mathcal{L}_{\text{fsl}}^{s} + \mathcal{L}_{\text{fsl}}^{t}. \quad (4)$$

2) Correlation Alignment: CORAL aims to align feature distributions between the source and target domains by minimizing the difference between their covariance matrices in Euclidean space. It is implemented as follows:

$$\mathcal{L}_{\text{coral}} = \frac{1}{4D^2}\left\|\mathbf{C}_s - \mathbf{C}_t\right\|_F^2. \quad (5)$$

In (5), $\|\cdot\|_F$ denotes the Frobenius norm, D is the feature dimension, and $\mathbf{C}_s$ and $\mathbf{C}_t$ represent the covariance matrices of the source and target domains, respectively.

D. AAM

As shown in Fig. 1, the AAM applies different augmentation strengths to target domain samples via a query encoder and a momentum-updated key encoder under asymmetric strategies: weak augmentation is used on query samples to preserve semantic consistency and feature stability, while strong augmentation on key samples enhances feature diversity and prototype robustness. This asymmetric design encourages the model to learn more discriminative and generalizable features through CL. Implementation details are given below.

1) *Asymmetric Augmentation Strategy:* Mask-based augmentation has recently shown strong performance in the RGB image domain [57], [58]. By masking a portion of pixels, the model is encouraged to learn more comprehensive representations. However, the fundamental differences between HSI and RGB images, including the number of bands, spectral correlations, and semantic structures, make direct transfer of these methods suboptimal. Therefore, PGSMC introduces tailored adjustments for HSI data, as detailed in Table III.
   1) *Spectral–Spatial Masking:* Unlike RGB images with three channels, HSI contains hundreds of correlated bands. Traditional spatial-only masking is inadequate for such data. To address this, we apply spectral–spatial masking, which jointly occludes spectral and spatial regions, encouraging the model to learn robust features from incomplete inputs.
   2) *Gaussian Noise With Varying Intensity:* HSI data exhibit strong interband correlation along with distinctive spectral signatures. Applying Gaussian noise with different intensities to the query and key views helps the model learn invariant and abstract representations, improving its ability to capture essential spectral patterns.
   3) *Preserving the Central Pixel:* While labels in RGB tasks are often image-level, HSIC focuses on pixel-level prediction. Thus, PGSMC preserves the central pixel during masking to retain the core semantic information needed for accurate classification.
2) *Query Encoder:* The query encoder shares parameters with the residual feature extractor in the EM. For each labeled pixel in the target domain, a patch centered on that pixel is extracted to form the original view $\mathbf{v} \in \mathbb{R}^{7\times7\times100}$. The query view $\mathbf{v}^q$, obtained via weak augmentation, is then fed into the query encoder f_q to generate its embedded representation $\mathbf{v} = f_q(\mathbf{v}^q)$.

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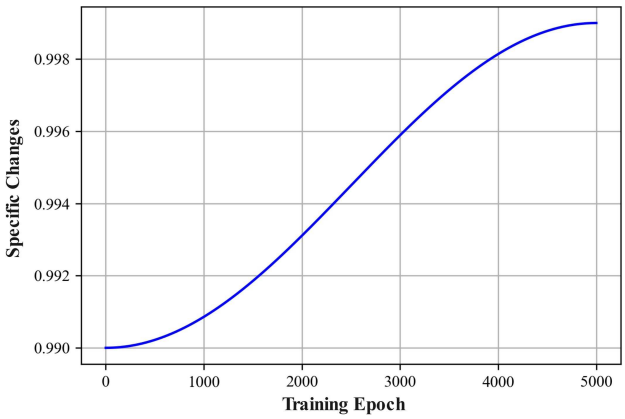


Fig. 3. Specific changes in the m_{step} value during training. Cosine scheduling sets a smaller momentum coefficient at the early stage (faster updates) and approaches 1 at the later stage (slower updates), facilitating stable feature alignment throughout training.

3) *Key Encoder:* The key encoder is not updated via backpropagation. Instead, it follows momentum-based parameter updates derived from the query encoder during training, which can be formulated as follows:

$$\boldsymbol{\theta}_k \leftarrow m \cdot \boldsymbol{\theta}_k + (1-m) \cdot \boldsymbol{\theta}_q. \tag{6}$$

Here, $\boldsymbol{\theta}_k$ and $\boldsymbol{\theta}_q$ denote the parameters of the key encoder and query encoder, respectively. The momentum coefficient m controls the degree to which past parameters are retained. To ensure stable feature alignment and representative prototype estimation, PGSMC employs a cosine schedule to dynamically adjust the parameter m throughout the training process

$$m_{\text{step}} = m_{\text{final}} - (m_{\text{final}} - m_{\text{base}}) \cdot \frac{1 + \cos\left(\pi \cdot \frac{\text{step}}{\text{max_steps}}\right)}{2}. \tag{7}$$

In (7), m_{step} denotes the momentum coefficient at the current training step; m_{base} and m_{final} are the initial and final momentum values (set to 0.99 and 0.999, respectively); step is the current epoch, and max_steps is the total number of training epochs. The adjustment curve is shown in Fig. 3. The key view $\mathbf{v}^k$, obtained by applying strong augmentation to the original view $\mathbf{v}$, is then fed into the key encoder f_k to produce the feature representation $\mathbf{k} = f_k(\mathbf{v}^k)$.

E. MPCM

Although CORAL reduces domain shifts, its rigid global alignment may distort class structures and blur decision boundaries. To address this, MPs are adopted in PGSMC as contrastive targets, promoting global class aggregation and clearer boundaries. An MQ is further employed to incorporate historical features, enhancing prototype consistency and mitigating insufficient supervision in few-shot settings. It is worth noting that CL is applied only to the target domain, as aligning prototypes with the target distribution is crucial for mitigating class shift, whereas CL on the source domain would bias the representations toward irrelevant source-specific features [59], [60]. The implementation is detailed as follows.

The key feature $\mathbf{k}_j$ and corresponding label y_j generated by the AAM are stored in the MQs $\mathbf{Q}$ and $\mathbf{Y}$, following the update rules defined in the following equations:

$$\mathbf{Q}_{[:,(p+j) \bmod K]} \leftarrow \mathbf{k}_j^T \quad \forall j = 0, 1, \ldots, B-1 \tag{8}$$

$$\mathbf{Y}_{[:,(p+j) \bmod K]} \leftarrow y_j \quad \forall j = 0, 1, \ldots, B-1 \tag{9}$$

where p denotes the queue pointer indicating the next write position, and $(p + i)$ mod K ensures circular writing when the queue tail lacks sufficient space. For each class $c \in C$, the initial MP is computed as the mean vector of all its corresponding samples from the queue $\mathbf{Q}$. $\widehat{\mathbf{p}}_c^{(t)}$ denotes the MP of class c at the t-th training step (with $t = 0$ at initialization), which is formally defined as follows:

$$\widehat{\mathbf{p}}_c^{(t)} = \frac{1}{|\mathcal{K}_c|} \sum_{\mathbf{k}_j \in \mathcal{K}_c} \mathbf{k}_j. \tag{10}$$

Here, $\mathcal{K}_c$ denotes the set of $\mathbf{k}$ belonging to class c. During training, $\widehat{\mathbf{p}}_c^{(t)}$ is updated as follows:

$$\widehat{\mathbf{p}}_c^{(t)} = m \cdot \widehat{\mathbf{p}}_c^{(t-1)} + (1-m) \cdot \mathbf{p}_c^{(t)} \tag{11}$$

where $\widehat{\mathbf{p}}_c^{(t-1)}$ denotes the MP from the previous step, and $\mathbf{p}_c^{(t)}$ is the prototype of class c computed from the current batch. Then, for each query feature $\mathbf{q}_i$, its distance to $\widehat{\mathbf{p}}_c^{(t)}$ is computed

$$\mathbf{z}_{i,c} = -\frac{1}{T} \left\| \mathbf{q}_i - \widehat{\mathbf{p}}_c^{(t)} \right\|^2. \tag{12}$$

In (12), T denotes the temperature scaling factor, set to 0.1 in this work. A supervision mask $\mathbf{M}_{i,c}$ is then defined to identify positive samples

$$\mathbf{M}_{i,c} = \begin{cases} 1, & \text{if } y_i = c \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where y_i denotes the label of the query feature $\mathbf{q}_i$. The final MP contrastive (MPC) loss is defined as follows:

$$\mathcal{L}_{\text{mpc}} = -\frac{1}{B} \sum_{i=1}^{B} \log \left(\frac{\sum_{c \in \mathcal{C}} \mathbf{M}_{i,c} \cdot \exp\left(\mathbf{z}_{i,c}\right)}{\sum_{c \in \mathcal{C}} \exp\left(\mathbf{z}_{i,c}\right)} \right). \tag{14}$$

Here, $\mathcal{L}_{\text{mpc}}$ aligns query features with MPs, promoting intra-class compactness and interclass separability.

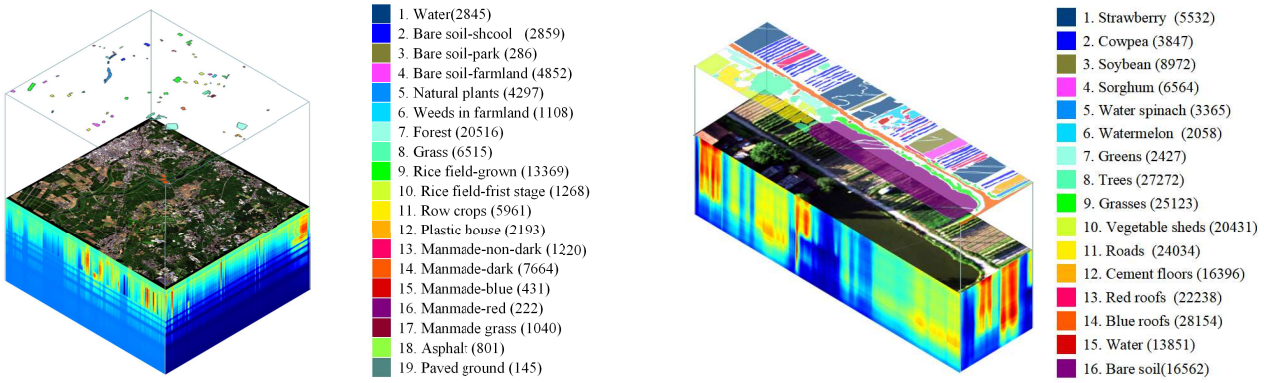
F. Total Loss

The total loss of PGSMC is composed of the FSL loss $\mathcal{L}_{\text{fsl}}$, CORAL loss $\mathcal{L}_{\text{coral}}$, and MPC loss $\mathcal{L}_{\text{mpc}}$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{fsl}} + \mathcal{L}_{\text{coral}} + \lambda \mathcal{L}_{\text{mpc}} \tag{15}$$

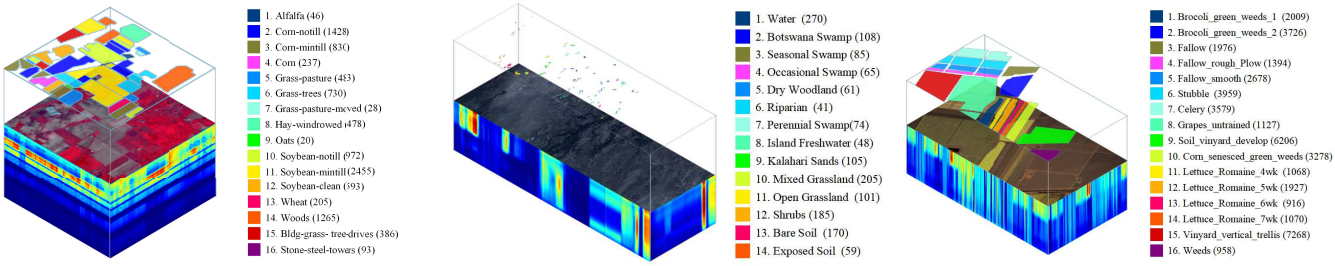
where $\mathcal{L}_{\text{fsl}}$ is used to learn similarity-based embedding spaces for both source and target domains, where samples from the same class are mapped closely, while those from different classes are well separated. $\mathcal{L}_{\text{coral}}$ explicitly reduces the distribution shift between domains to enable knowledge transfer. However, this global alignment often weakens interclass discriminability, resulting in blurred class boundaries. To address this, $\mathcal{L}_{\text{mpc}}$ constructs momentum class-level prototypes

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(a) (b)

Fig. 4. Source domain datasets used in the experiment. The land cover types and the number of samples for each dataset are indicated, respectively. (a) CK. (b) HC.



(a) (b) (c)

Fig. 5. Target domain datasets used in the experiment. The land cover types and the number of samples for each dataset are indicated, respectively. (a) IP. (b) BO. (c) SA.

in the target domain. By contrasting samples with their corresponding prototypes, intraclass compactness and interclass separability are explicitly enhanced, improving discrimination and generalization in the target domain. Since $\mathcal{L}_{\text{mpc}}$ is applied only to the target domain, a weighting factor λ is introduced and set to 2.0 in this work.

IV. Experiments

A. Datasets Introduction

To ensure consistency with the comparative methods, the Chikusei and WHU-Hi-HanChuan (HC) datasets were adopted as the source domains, while Indian Pines (IP), Botswana (BO), and Salinas datasets were used as the target domains for evaluating the performance of PGSMC. The details are as follows.

1) *Chikusei (CK):* The dataset is captured by the Headwall Hyperspec-VNIR-C sensor over Chikusei City, Japan. It consists of 2517 × 2335 pixels with a spatial resolution of 2.5 m. A total of 128 bands are provided, covering wavelengths from 343 to 1018 nm, with 19 classes. The false-color image and ground truth are shown in Fig. 4(a).
2) *WHU-Hi-HanChuan:* The dataset is collected by the Headwall Nano-Hyperspec imaging sensor on a Leica Aibot X6 UAV V1 platform over Hanchuan, Hubei province, China. It contains 1217 × 303 pixels with a spatial resolution of approximately 0.109 m. The dataset covers a spectral range from 400 to 1000 nm, comprising 274 bands and 16 classes. The false-color image and ground truth are shown in Fig. 4(b).
3) *Indian Pines:* This dataset is collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over Indiana, USA. It contains 145 × 145 pixels with a spatial resolution of approximately 20 m. After removing 20 water absorption bands (104–105, 150–163, and 220), 200 bands ranging from 400 to 2500 nm are used. The dataset includes 16 classes. The false-color image and ground truth are shown in Fig. 5(a).
4) *Botswana:* This dataset is acquired by NASA's EO-1 satellite over the Okavango Delta, BO. It has a size of 1476 × 256 pixels with a spatial resolution of approximately 20 m. Among the 242 spectral bands (400–2500 nm), 145 bands are used after removing noisy bands (1–9, 56–81, 98–101, 120–133, and 165–186). It contains 14 classes. The false-color image and ground truth are shown in Fig. 5(b).
5) *Salinas (SA):* This dataset is collected by the AVIRIS sensor over the Salinas Valley, California, USA. It consists of 512 × 217 pixels with a spatial resolution of approximately 3.7 m. A total of 204 bands covering 400–2500 nm are used, with 16 classes. The false-color image and ground truth are shown in Fig. 5(c).

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TABLE III
ASYMMETRIC AUGMENTATION STRATEGY

Name	Spatial Mask Ratio	Spectral Mask Ratio	Standard Deviation
Weak Augmentation	0.3	0.05	0.01
Strong Augmentation	0.6	0.1	0.02

B. Comparison Methods

To validate the effectiveness of PGSMC, comparisons are conducted with seven state-of-the-art methods in hyperspectral few-shot classification: DCFSL (IEEE TGRS, 2022) [31], HFSL (IEEE GRSL, 2022) [30], DM-MRN (IEEE TGRS, 2023) [24], FSCF-SSL (IEEE TGRS, 2023) [38], GCC-FSL (IEEE TGRS, 2024) [39], CTF-SSCL (IEEE TGRS, 2024) [40], and MLPA (IEEE TGRS, 2025) [34].

1) *DCFSL:* Few-shot classification and DA are jointly optimized via conditional adversarial learning within a unified architecture.
2) *HFSL:* Designed to transfer knowledge from the natural image domain, this approach fuses spectral and spatial cues to boost classification accuracy.
3) *DM-MRN:* Combines image-to-class and I2I relations in a multistage network, capturing multilevel features and enhancing relational reasoning.
4) *FSCF-SSL:* Integrates self-supervised learning and FSL by utilizing natural image pretraining and CL to extract transferable and class-invariant features.
5) *GCC-FSL:* Mitigates domain shift in few-shot HSIC by constructing positive–negative prototype pairs and leveraging a graph convolution contrast module to enhance cross-domain feature alignment.
6) *CTF-SSCL:* Enhances spectral–spatial discriminability by combining supervised contrastive loss with a novel spectral feature perturbation strategy within a cascaded CNN-Transformer framework.
7) *MLPA:* Employs multiscale feature alignment and prototype-based adversarial training to address domain shift and suppress negative transfer.

Among these methods, DM-MRN is not designed for cross-domain scenarios. DCFSL, GCC-FSL, CTF-SSCL, and MLPA are implemented with CK and HC datasets as the source domains, while HFSL and FSCF-SSL leverage mini-ImageNet for spectral–spatial feature transfer. The hyperparameters of all compared methods are kept consistent with those reported in the original publications.

C. Experimental Setting

All experiments are conducted on a server with PyTorch 2.2 and Python 3.11, running on an Intel i7-13700 CPU, an NVIDIA RTX 4090 GPU, and 32 GB RAM. In PGSMC, 7×7 image patches are used as input. The learning rate is 0.005, and parameters are updated using the Adam optimizer for 5000 iterations. For each class, one–five labeled target samples are selected. As DM-MRN cannot handle single-sample classification, Gaussian noise is added for data augmentation in such cases. For DCFSL, HFSL, FSCF-SSL, GCC-FSL, CTF-SSCL, and MLPA, 200 labeled source samples per class are randomly selected to learn transferable knowledge. In the target domain, one augmented sample per class is used as the support set, and the rest form the query set. Prototypes are computed from the support set, and distances are measured from the query set. All samples are normalized to follow a standard normal distribution. Performance is evaluated using class accuracy (CA), overall accuracy (OA), average accuracy (AA), and the Kappa coefficient $\times 100$ (κ). Each experiment is repeated ten times to reduce sampling bias. In addition, classification maps are visualized for qualitative analysis.

D. Experimental Results and Analysis

Tables IV–VI report the classification performance of all methods with CK as the source domain and IP, BO, and SA as the target domains, using only five labeled samples per class. The best results are highlighted in bold for clarity. As shown in Table IV, PGSMC achieves the best performance on the IP dataset, with an OA of 80.28%, an AA of 88.38%, and a κ of 77.71, significantly outperforming instance-level contrastive methods such as FSCF-SSL and CTF-SSCL. Moreover, PGSMC attains the best CA in 10 out of 16 classes, demonstrating strong class modeling capability. Notably, PGSMC was the only method among the eight evaluated to achieve an OA exceeding 80%, demonstrating the effectiveness of its class-level MPC approach in preserving class structure. On the BO dataset (Table V), PGSMC again achieves leading performance, with OA, AA, and κ reaching 97.72%, 97.71%, and 97.74, respectively. This outperforms the prototype-based DA method MLPA by approximately 2% in OA. This result highlights the contribution of the MQ mechanism to stable prototype estimation. Similarly, Table VI presents the results on the SA dataset, where PGSMC maintains top performance across all three metrics, achieving 93.02% OA, 95.86% AA, and 92.22 κ. Compared to the state-of-the-art CL methods GCC-FSL and CTF-SSCL, PGSMC demonstrates significant improvements in OA, AA, and κ values. Overall, these results confirm that the prototype-guided supervised contrastive strategy in PGSMC effectively captures intraclass relationships and enhances class boundaries, leading to improved classification accuracy and discriminative capability.

To provide a more intuitive comparison of classification performance across different target domains, Figs. 5–7 present the classification results of all methods using only five labeled samples per class. As shown in Fig. 6, on the IP dataset, PGSMC demonstrates strong discriminative ability in classifying class 2 and class 10, while other methods generally fail to distinguish among classes 2, 10, and 11. This highlights the superior capability of PGSMC in handling interclass similarity. In Fig. 7, the results on the BO dataset indicate that the classification by PGSMC is highly consistent with the ground-truth land cover distribution, demonstrating excellent spatial coherence. In contrast, other methods exhibit noticeable confusion, particularly in classes 5, 6, 8, and 9, indicating limited class separability. These findings further validate the robustness and adaptability of the proposed momentum-based

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TABLE IV

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF IP DATASET

Class	DCFSL	HFSL	DM-MRN	FSCF-SSL	GCC-FSL	CTF-SSCL	MLPA	PGSMC
1	94.15 ± 7.33	99.51 ± 0.98	99.02 ± 2.24	**99.51 ± 0.98**	88.54 ± 15.54	96.83 ± 5.78	92.20 ± 11.94	98.78 ± 2.94
2	47.48 ± 12.01	54.10 ± 11.53	64.45 ± 9.29	59.16 ± 11.24	43.18 ± 8.15	54.26 ± 13.26	50.51 ± 11.08	**67.38 ± 15.58**
3	52.69 ± 8.68	55.98 ± 13.67	45.25 ± 11.41	65.25 ± 13.64	51.56 ± 8.41	59.37 ± 7.54	50.69 ± 12.59	**75.42 ± 6.57**
4	78.06 ± 9.52	90.78 ± 12.87	90.00 ± 6.38	92.28 ± 11.50	76.81 ± 11.48	87.07 ± 11.37	82.41 ± 6.00	**96.12 ± 2.69**
5	70.84 ± 10.29	68.12 ± 12.68	73.74 ± 6.44	70.94 ± 12.17	73.62 ± 7.61	77.15 ± 10.82	72.24 ± 12.19	**81.11 ± 7.33**
6	86.50 ± 4.80	74.90 ± 14.07	92.79 ± 2.41	85.12 ± 6.54	81.90 ± 7.43	87.75 ± 6.09	87.20 ± 6.19	**97.48 ± 1.54**
7	99.13 ± 1.74	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	97.39 ± 3.98	**100.00 ± 0.00**	99.57 ± 1.30	**100.00 ± 0.00**
8	83.51 ± 11.89	**99.92 ± 0.19**	98.44 ± 1.20	99.83 ± 0.39	88.75 ± 7.83	93.78 ± 10.40	82.62 ± 12.68	95.94 ± 9.11
9	99.33 ± 2.00	99.33 ± 2.00	**100.00 ± 0.00**	**100.00 ± 0.00**	94.00 ± 15.90	98.67 ± 2.67	**100.00 ± 0.00**	**100.00 ± 0.00**
10	67.44 ± 7.21	67.37 ± 13.11	61.25 ± 15.53	66.63 ± 10.74	61.22 ± 8.73	64.06 ± 11.13	66.06 ± 9.94	**76.14 ± 10.37**
11	58.64 ± 11.66	74.95 ± 7.44	54.56 ± 9.62	**76.04 ± 8.94**	58.69 ± 12.31	65.87 ± 13.85	57.64 ± 12.64	75.40 ± 8.35
12	40.97 ± 6.50	67.28 ± 10.78	53.33 ± 11.74	71.41 ± 16.33	43.30 ± 10.98	47.82 ± 11.09	46.00 ± 10.46	**76.84 ± 14.48**
13	98.10 ± 2.81	98.45 ± 2.44	98.55 ± 2.09	99.75 ± 0.46	99.10 ± 1.39	99.10 ± 1.70	97.85 ± 3.61	**99.90 ± 0.20**
14	84.33 ± 9.55	91.57 ± 6.84	88.77 ± 6.35	**92.75 ± 4.51**	85.62 ± 7.14	87.48 ± 5.40	87.33 ± 6.99	84.65 ± 8.53
15	73.70 ± 8.93	**91.00 ± 9.76**	87.38 ± 9.65	90.47 ± 10.47	74.62 ± 10.05	82.20 ± 9.77	77.43 ± 9.44	90.31 ± 9.50
16	**99.43 ± 1.37**	97.50 ± 3.69	96.70 ± 6.81	95.45 ± 6.06	98.07 ± 2.04	99.77 ± 0.45	98.52 ± 1.76	98.64 ± 2.02
OA (%)	65.74 ± 2.57	74.03 ± 2.71	69.28 ± 3.30	76.96 ± 2.74	64.79 ± 2.58	70.86 ± 3.21	66.59 ± 2.81	**80.28 ± 3.26**
AA (%)	77.14 ± 1.70	83.17 ± 2.25	81.51 ± 1.84	85.29 ± 2.60	76.02 ± 2.07	81.32 ± 2.28	78.02 ± 1.43	**88.38 ± 1.80**
κ	61.47 ± 2.66	70.64 ± 2.98	65.66 ± 3.61	73.87 ± 3.06	60.30 ± 2.64	67.10 ± 3.49	62.43 ± 2.99	**77.71 ± 3.67**

TABLE V

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF BO DATASET

Class	DCFSL	HFSL	DM-MRN	FSCF-SSL	GCC-FSL	CTF-SSCL	MLPA	PGSMC
1	99.77 ± 0.68	**100.00 ± 0.00**	94.75 ± 6.43	**100.00 ± 0.00**	99.77 ± 0.56	99.66 ± 0.78	99.85 ± 0.35	99.81 ± 0.45
2	**100.00 ± 0.00**	96.98 ± 9.06	99.90 ± 0.31	89.27 ± 13.98	99.27 ± 1.48	99.48 ± 1.07	99.38 ± 1.06	**100.00 ± 0.00**
3	**98.41 ± 1.70**	97.15 ± 4.02	97.85 ± 3.12	97.56 ± 4.02	97.97 ± 2.06	96.91 ± 4.34	95.89 ± 6.22	97.89 ± 4.50
4	**99.33 ± 0.86**	91.00 ± 12.61	**99.86 ± 0.22**	93.76 ± 10.41	97.00 ± 3.09	98.19 ± 2.39	98.10 ± 3.36	98.86 ± 1.90
5	86.40 ± 7.93	84.77 ± 9.19	75.34 ± 10.60	90.64 ± 6.82	81.97 ± 7.52	82.16 ± 7.39	82.69 ± 7.41	**94.77 ± 5.27**
6	89.85 ± 8.64	86.44 ± 8.61	89.77 ± 6.78	83.48 ± 11.89	93.22 ± 6.67	91.44 ± 8.64	**94.28 ± 8.13**	93.07 ± 10.08
7	99.17 ± 1.37	95.47 ± 5.70	**100.00 ± 0.00**	97.05 ± 3.33	99.29 ± 0.70	98.82 ± 2.54	99.69 ± 0.55	**100.00 ± 0.00**
8	**99.75 ± 0.76**	93.74 ± 11.44	91.16 ± 11.38	98.18 ± 5.45	99.70 ± 0.76	98.23 ± 4.50	98.69 ± 2.71	99.34 ± 1.97
9	96.47 ± 3.76	93.75 ± 6.07	91.04 ± 9.23	95.08 ± 7.09	95.89 ± 3.92	**96.54 ± 3.74**	96.25 ± 3.48	96.41 ± 5.20
10	99.55 ± 0.72	99.26 ± 1.65	99.88 ± 0.26	99.71 ± 0.74	99.59 ± 0.86	98.89 ± 1.59	99.84 ± 0.33	**100.00 ± 0.00**
11	99.17 ± 1.05	96.33 ± 5.17	85.83 ± 10.04	**99.57 ± 0.68**	99.07 ± 0.87	96.73 ± 4.36	98.37 ± 1.93	98.73 ± 1.62
12	99.49 ± 0.54	99.94 ± 0.17	93.07 ± 9.06	98.69 ± 3.23	98.58 ± 2.68	99.32 ± 0.84	98.75 ± 1.78	**100.00 ± 0.00**
13	97.76 ± 3.49	97.26 ± 4.80	91.90 ± 9.69	99.73 ± 0.54	97.53 ± 2.39	99.09 ± 1.79	99.24 ± 1.55	**100.00 ± 0.00**
14	93.22 ± 7.28	94.67 ± 6.88	**96.11 ± 7.24**	95.33 ± 6.32	85.11 ± 8.88	89.11 ± 6.57	93.33 ± 6.46	89.11 ± 8.46
OA (%)	96.90 ± 1.03	94.56 ± 1.61	92.55 ± 1.79	95.81 ± 1.43	96.24 ± 0.87	96.05 ± 1.32	96.65 ± 1.04	**97.92 ± 1.34**
AA (%)	97.02 ± 1.13	94.77 ± 1.33	93.32 ± 1.76	95.58 ± 1.46	96.00 ± 1.01	96.04 ± 1.41	96.74 ± 1.08	**97.71 ± 1.44**
κ	96.63 ± 1.12	94.10 ± 1.73	91.93 ± 1.94	95.45 ± 1.55	95.93 ± 0.94	95.71 ± 1.43	96.37 ± 1.12	**97.74 ± 1.45**

TABLE VI

CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF SA DATASET

Class	DCFSL	HFSL	DM-MRN	FSCF-SSL	GCC-FSL	CTF-SSCL	MLPA	PGSMC
1	99.01 ± 1.21	97.01 ± 7.78	99.28 ± 0.82	**99.98 ± 0.05**	98.26 ± 1.79	99.54 ± 0.78	98.93 ± 0.89	99.77 ± 0.28
2	**99.77 ± 0.23**	93.13 ± 9.21	99.55 ± 1.11	97.22 ± 6.29	98.87 ± 1.27	99.64 ± 0.49	99.15 ± 1.34	99.25 ± 1.64
3	95.63 ± 4.86	95.63 ± 6.21	**100.00 ± 0.00**	97.97 ± 2.97	93.80 ± 4.79	91.34 ± 10.43	93.84 ± 4.32	95.11 ± 4.83
4	99.73 ± 0.27	99.22 ± 1.41	99.66 ± 0.30	98.98 ± 2.54	99.42 ± 0.91	99.40 ± 0.64	99.19 ± 1.28	**99.79 ± 0.23**
5	90.17 ± 2.41	95.42 ± 5.20	97.18 ± 3.04	**97.86 ± 0.71**	90.91 ± 4.00	93.50 ± 3.82	91.57 ± 3.10	95.86 ± 2.45
6	98.52 ± 1.62	98.52 ± 2.62	96.90 ± 3.54	99.81 ± 0.23	99.18 ± 0.95	99.46 ± 0.63	99.67 ± 0.56	**99.82 ± 0.21**
7	99.41 ± 0.57	98.63 ± 1.18	**99.82 ± 0.27**	99.61 ± 0.53	98.98 ± 1.23	99.36 ± 0.98	99.05 ± 0.81	99.80 ± 0.27
8	77.93 ± 4.22	76.82 ± 12.17	75.02 ± 13.53	76.43 ± 9.65	79.25 ± 6.91	76.38 ± 7.10	80.93 ± 3.95	**86.50 ± 4.15**
9	99.45 ± 0.84	99.09 ± 1.27	**99.79 ± 0.30**	99.65 ± 0.42	98.54 ± 1.60	98.91 ± 1.38	98.82 ± 1.30	99.64 ± 0.68
10	84.53 ± 3.81	94.03 ± 3.68	89.92 ± 8.51	**94.99 ± 4.79**	81.53 ± 7.09	84.99 ± 7.20	85.26 ± 5.45	88.22 ± 4.57
11	99.16 ± 0.60	98.85 ± 1.38	99.31 ± 1.32	99.01 ± 1.93	95.09 ± 3.85	97.42 ± 2.22	97.67 ± 1.98	**99.49 ± 0.77**
12	99.65 ± 0.43	98.36 ± 2.23	94.42 ± 5.97	97.21 ± 5.35	98.36 ± 2.74	99.59 ± 0.48	98.62 ± 2.27	**99.98 ± 0.05**
13	99.13 ± 0.94	98.69 ± 1.06	96.20 ± 2.85	**99.45 ± 0.65**	97.45 ± 3.52	98.94 ± 1.27	99.15 ± 0.90	98.83 ± 1.01
14	97.81 ± 1.89	95.88 ± 3.53	93.56 ± 5.80	97.38 ± 2.52	97.06 ± 3.52	98.24 ± 2.88	**98.85 ± 0.54**	98.69 ± 1.27
15	77.17 ± 7.14	69.94 ± 19.36	**84.00 ± 6.54**	77.31 ± 10.04	77.80 ± 4.86	77.72 ± 6.82	72.35 ± 7.98	80.38 ± 8.15
16	91.73 ± 4.17	95.57 ± 4.97	**97.64 ± 2.36**	96.60 ± 4.60	89.27 ± 7.39	94.26 ± 4.09	89.25 ± 5.58	92.56 ± 5.25
OA (%)	90.12 ± 1.22	89.14 ± 2.09	91.08 ± 2.39	90.96 ± 2.06	89.83 ± 1.15	89.96 ± 1.43	89.94 ± 1.30	**93.02 ± 0.89**
AA (%)	94.30 ± 0.60	94.05 ± 1.49	95.14 ± 1.09	95.59 ± 1.09	93.36 ± 0.99	94.29 ± 0.99	93.89 ± 0.96	**95.86 ± 0.54**
κ	89.01 ± 1.35	87.93 ± 2.31	90.11 ± 2.62	89.96 ± 2.26	88.70 ± 1.28	88.86 ± 1.57	88.81 ± 1.44	**92.22 ± 0.99**

prototype contrastive architecture in few-shot settings. Similarly, Fig. 8 shows that PGSMC achieves the highest CA in classes 8 and 15 on the SA dataset. Although DM-MRN yields comparable performance in class 15, it fails to preserve object boundaries. Other methods also show significant confusion between classes 8 and 15. Overall, PGSMC not

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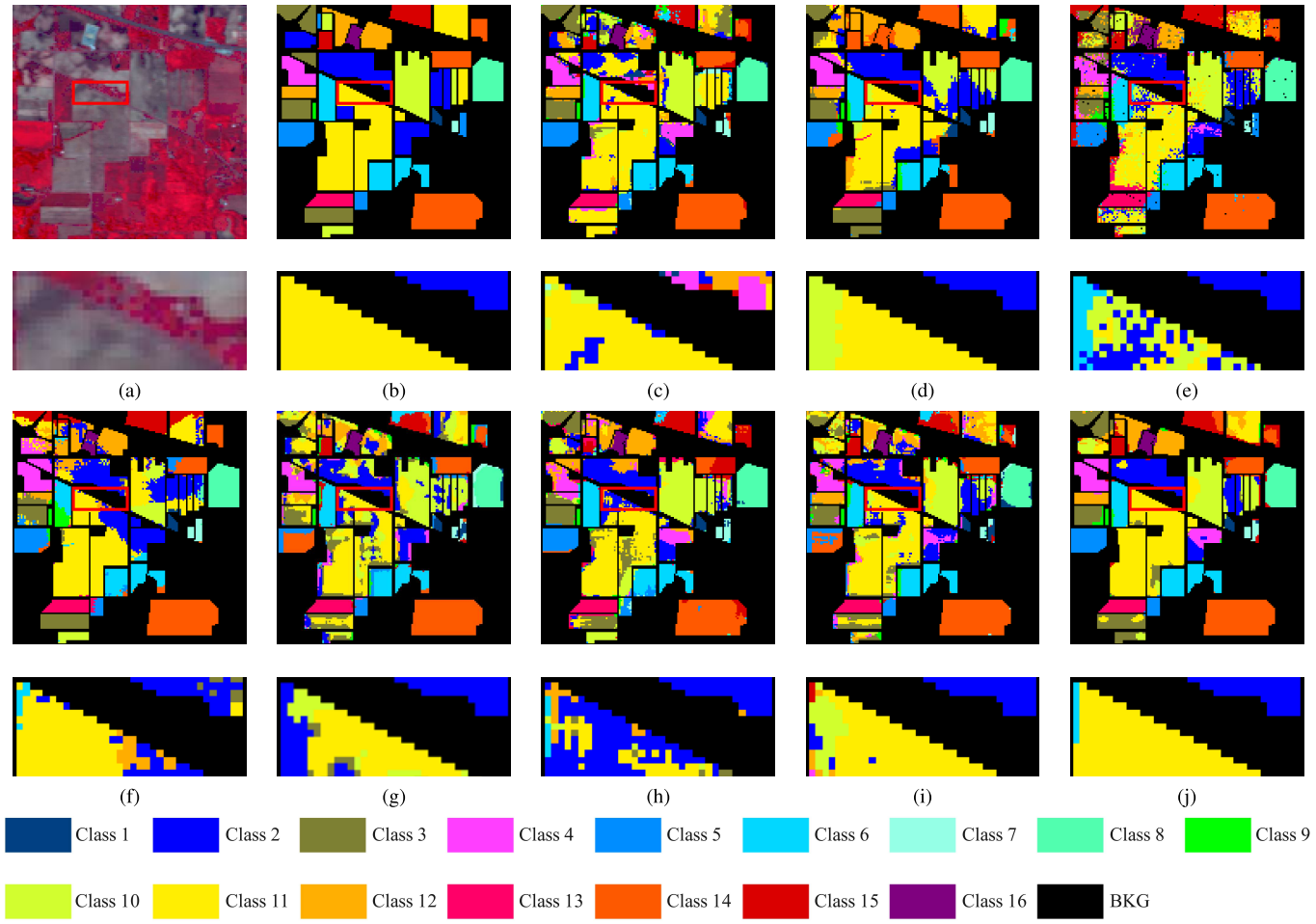


Fig. 6. Classification maps of IP dataset. (a) False-color image. (b) Ground-truth map. (c) DCFSL. (d) HFSL. (e) DM-MRN. (f) FSCF-SSL. (g) GCC-FSL. (h) CTF-SSCL. (i) MLPA. (j) PGSMC. Best zoomed-in view.

TABLE VII
CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF HC AS THE SOURCE DOMAIN

Dataset	Metric	DCFSL	HFSL	DM-MRN	FSCF-SSL	GCC-FSL	CTF-SSCL	MLPA	PGSMC
IP	OA	69.53 ± 2.85	74.03 ± 2.71	69.28 ± 3.30	76.96 ± 2.74	67.09 ± 0.93	70.59 ± 4.22	68.13 ± 3.21	**79.77 ± 2.60**
	AA	80.72 ± 1.56	83.17 ± 2.25	81.51 ± 1.84	85.29 ± 2.60	77.69 ± 1.42	81.69 ± 1.75	79.61 ± 1.84	**88.10 ± 1.41**
	κ	65.77 ± 3.19	70.64 ± 2.98	65.66 ± 3.61	73.87 ± 3.06	63.00 ± 1.06	66.96 ± 4.51	64.24 ± 3.50	**77.15 ± 2.93**
BO	OA	97.15 ± 0.83	94.56 ± 1.61	92.55 ± 1.79	95.81 ± 1.43	96.43 ± 0.84	95.34 ± 1.40	96.38 ± 1.13	**97.89 ± 1.39**
	AA	97.06 ± 0.96	94.77 ± 1.33	93.32 ± 1.76	95.58 ± 1.46	96.45 ± 0.88	95.60 ± 1.36	96.31 ± 1.18	**97.76 ± 1.56**
	κ	96.91 ± 0.90	94.10 ± 1.73	91.93 ± 1.94	95.45 ± 1.55	96.13 ± 0.91	94.95 ± 1.51	96.08 ± 1.22	**97.71 ± 1.50**
SA	OA	91.06 ± 0.99	89.14 ± 2.09	91.08 ± 2.39	90.96 ± 2.06	89.90 ± 0.90	89.79 ± 1.70	90.99 ± 1.07	**92.71 ± 0.82**
	AA	94.95 ± 0.62	94.05 ± 1.49	95.14 ± 1.09	95.59 ± 1.09	93.65 ± 0.65	93.82 ± 1.01	94.61 ± 0.85	**95.69 ± 0.43**
	κ	90.07 ± 1.10	87.93 ± 2.31	90.11 ± 2.62	89.96 ± 2.26	88.77 ± 1.00	88.66 ± 1.89	89.98 ± 1.18	**91.89 ± 0.91**

only exhibits strong discriminative power in complex regions but also demonstrates clear advantages in preserving spatial boundary information.

To further assess the performance of PGSMC under extremely limited samples, OA, AA, and κ are compared across one–five labeled samples per class on different target domains. Fig. 9(a) shows the results on the IP dataset. PGSMC consistently outperforms all methods across all metrics, with particularly notable gains under the one-shot setting. The IP dataset features highly imbalanced class distributions, making it difficult to achieve high AA with few samples. Despite this, PGSMC leads FSCF-SSL by 8.51%, 6.66%, 5.29%, 3.58%, and 3.09% in AA under the one–five shot settings, confirming its robustness and superiority in rare class scenarios. As shown in Fig. 9(b), PGSMC also achieves the best results on the BO dataset, with all curves remaining above those of competing methods. In the one-shot setting, the proposed method outperforms the state-of-the-art CL method GCC-FSL by 8.78% (OA), 7.99% (AA), and 12.17 (κ), demonstrating its strong generalization ability under low-resource conditions. Fig. 9(c) further confirms PGSMC's advantage on the SA dataset, surpassing CTF-SSCL by 6.03%, 5.20%, 4.81%, 3.45%, and 3.06% in OA across one–five shots. In summary, the class-level supervised contrastive method in PGSMC enables more

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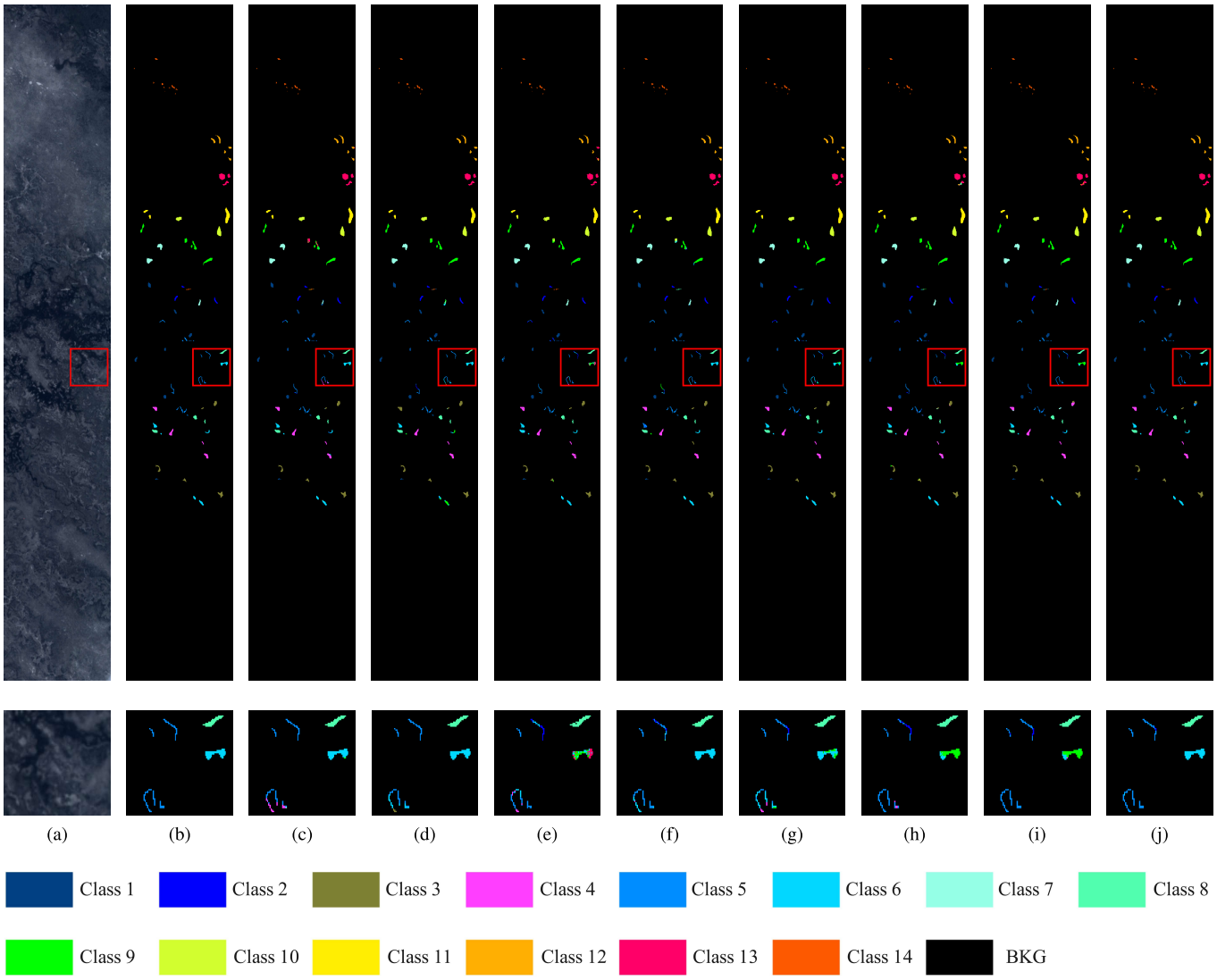


Fig. 7. Classification maps of BO dataset. (a) False-color image. (b) Ground-truth map. (c) DCFSL. (d) HFSL. (e) DM-MRN. (f) FSCF-SSL. (g) GCC-FSL. (h) CTF-SSCL. (i) MLPA. (j) PGSMC. Best zoomed-in view.

effective semantic modeling than its instance-level counterpart, leading to superior discriminative feature learning and ultimately higher performance in few-shot scenarios.

E. Multisource Domain Generalization Analysis

To further validate the generalization of PGSMC in multisource domain scenarios, the HC dataset, which exhibits significant differences from the target domain, was utilized as the source domain. The classification performance of all methods was verified using five samples, rendering these experiments more meaningful for the evaluation of cross-domain generalization capabilities. The specific classification results are presented in Table VII. Specifically, PGSMC achieves the best OA, AA, and Kappa coefficient across all datasets, consistently surpassing existing methods. In the IP dataset, PGSMC improves OA by 2.81% compared with FSCF-SSL and by 5.74% compared with HFSL. In the BO dataset, PGSMC achieves 97.89% OA, outperforming GCC-FSL (96.43%) and CTF-SSCL (95.34%). In the SA dataset, PGSMC again secures the highest OA of 92.71%. Overall, PGSMC maintains its superiority across multiple source–target combinations, demonstrating that the proposed prototype-driven momentum contrast mechanism can effectively adapt to diverse domain pairs and achieve accurate alignment of target-domain class semantics.

F. Visual Analysis of Feature Distribution

To further evaluate the feature separability and the preservation of class structures in PGSMC, Fig. 10 presents t-distributed stochastic neighbor embedding (t-SNE) visualizations of three recent FSL methods (FSCF-SSL, CTF-SSCL, and MLPA) together with PGSMC across different datasets. On the IP dataset, FSCF-SSL and CTF-SSCL, both based on instance-level CL, show similar distributions. Most clusters are poorly formed, with severe confusion observed among classes 2, 3, 10, and 11. MLPA, which adopts prototype alignment, preserves class structure to some extent, but overlaps persist among classes 2, 10, and 11. In contrast, PGSMC achieves

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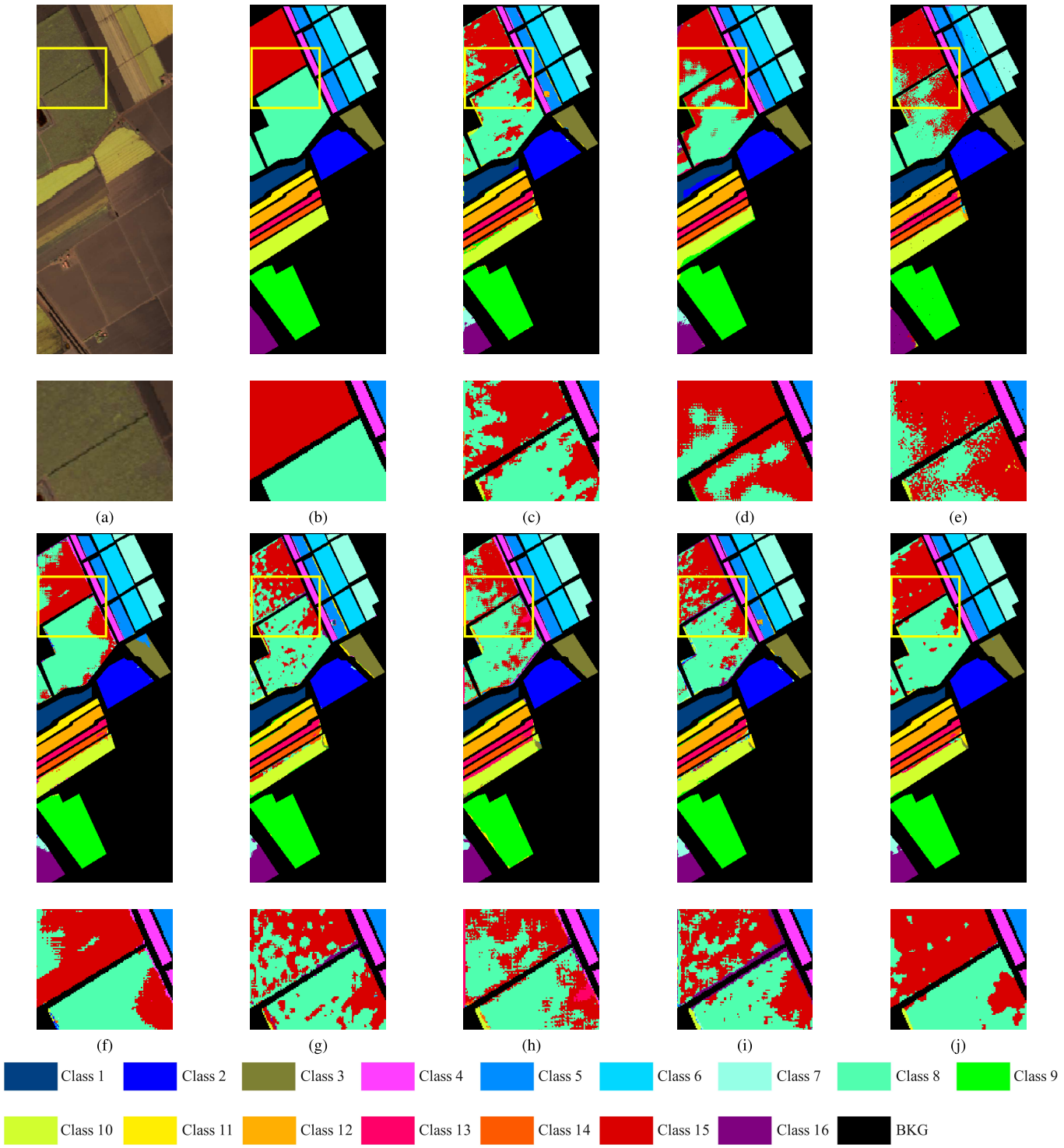


Fig. 8. Classification maps of SA dataset. (a) False-color image. (b) Ground-truth map. (c) DCFSL. (d) HFSL. (e) DM-MRN. (f) FSCF-SSL. (g) GCC-FSL. (h) CTF-SSCL. (i) MLPA. (j) PGSMC. Best zoomed-in view.

clearer and more distinct class boundaries through MPC learning. On the BO dataset, where all 14 classes show distinct spectral characteristics, the overall class structure is generally preserved by all methods. However, FSCF-SSL, CTF-SSCL, and MLPA still display overlapping points among classes 5, 6, and 14. PGSMC maintains clean, nonoverlapping boundaries, demonstrating its robustness in separating marginal samples. For the SA dataset, high spectral similarity and spatial proximity between classes 8 and 15 result in complete failure of separation by FSCF-SSL and CTF-SSCL. MLPA offers partial improvement, yet notable entanglement remains. PGSMC yields fewer overlaps and more coherent clusters, highlighting its superior ability to discriminate between highly similar classes. In summary, PGSMC not only improves feature-level

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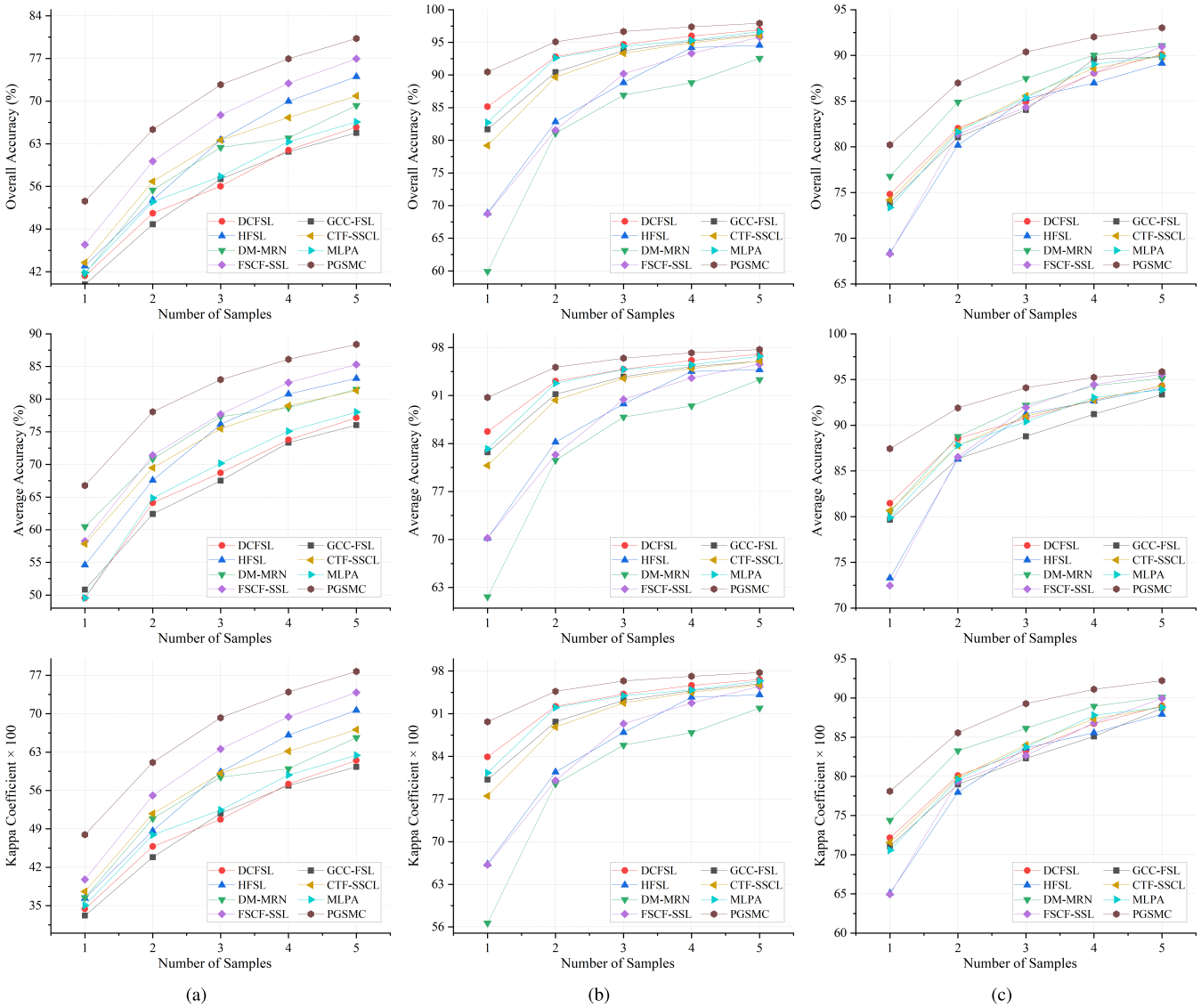


Fig. 9. Classification results of different datasets with one–five labeled samples. (a) IP. (b) BO. (c) SA.

separability but also better preserves class structure compared to existing approaches.

G. Hyperparameter Analysis

To further evaluate the impact of the momentum coefficient on model performance, a systematic average OA (AOA) assessment is conducted on the IP, BO, and SA datasets. Both fixed momentum values and a cosine scheduling strategy for dynamic momentum adjustment are investigated. The results are reported in Table VIII. Specifically, we obtain the following.

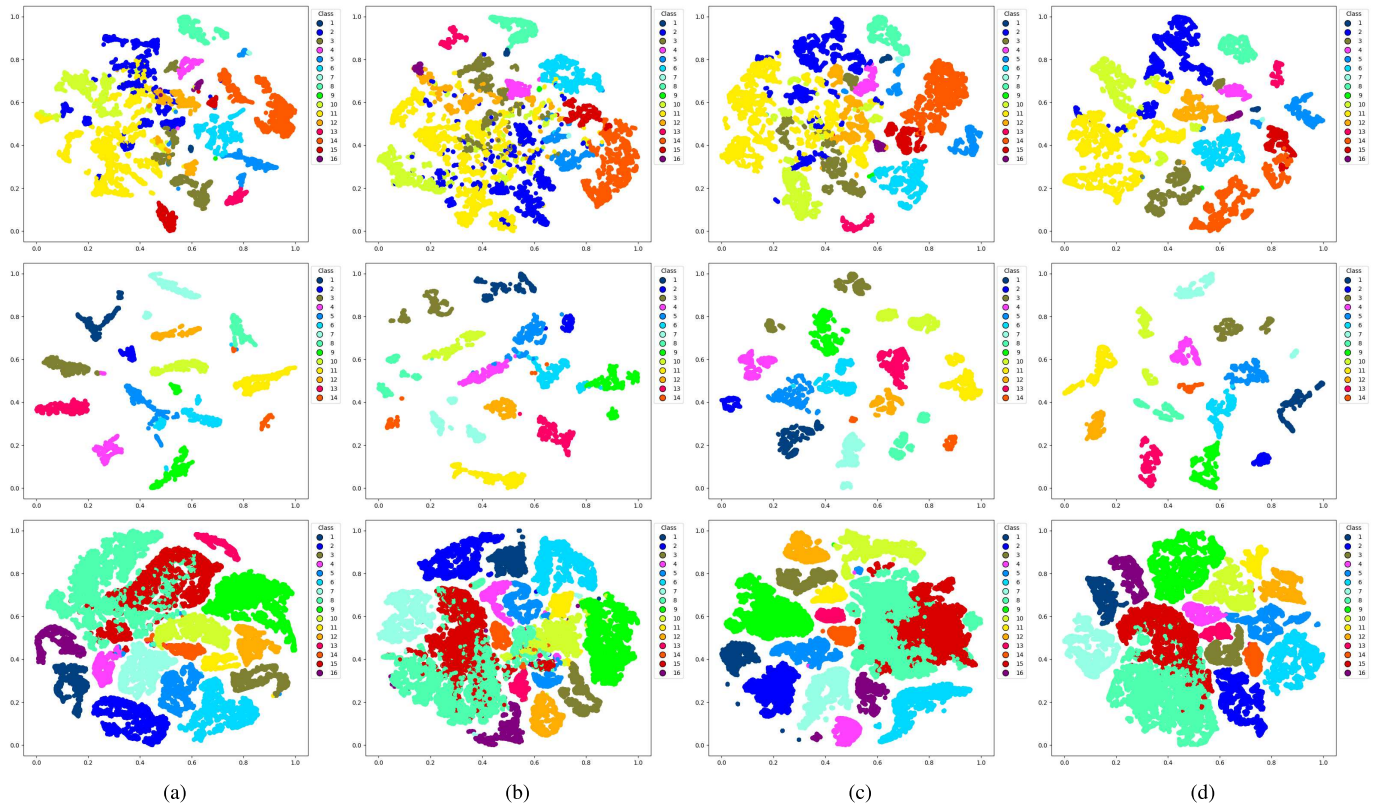
1) When $m = 0.9$, the worst performance is observed across all datasets. This is attributed to the low momentum coefficient being more susceptible to gradient fluctuations, resulting in unstable update directions.
2) When $m = 0.99$, although improved early-stage exploration is achieved, the model remains overly sensitive to gradient changes in later stages, hindering convergence to the optimal solution.
3) When $m = 0.999$, the large momentum coefficient typically results in smoother gradient updates but increases the accumulation of historical gradients, which may suppress responsiveness to current gradients.
4) When cosine scheduling is applied, the best performance is obtained across all datasets. This is because early-stage low momentum improves sensitivity to current gradients, while gradually increasing momentum stabilizes the optimization and promotes robust convergence.

H. Ablation Experiment

To evaluate the contributions of each core component in the proposed PGSMC, two sets of ablation experiments are conducted: 1) loss ablation and 2) module ablation.

1) *Loss Ablation Experiments:* To further evaluate the role of different loss functions, four loss combinations are designed based on the integration of CORAL for global feature alignment. Three contrastive objectives were sequentially introduced: the self-supervised contrastive

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(a) (b) (c) (d)

Fig. 10. Feature separation visualization using recent FSL methods across different datasets. The order of datasets applied from top to bottom are IP, BO, and SA. (a) FSCF-SSL. (b) CTF-SSCL. (c) MLPA. (d) PGSMC.

loss ($\mathcal{L}_{\text{cl}}$), the supervised contrastive loss ($\mathcal{L}_{\text{scl}}$), and the MPC loss ($\mathcal{L}_{\text{mpc}}$). For each objective, the balancing hyperparameter λ was uniformly set to 2.0. A comprehensive evaluation is conducted on all three datasets, with results summarized in Table IX.

1) $\mathcal{L}_{\text{fsl}}+\mathcal{L}_{\text{coral}}$: This baseline combination incorporates CORAL into the conventional FSL objective to align source and target distributions.
2) $\mathcal{L}_{\text{fsl}} + \mathcal{L}_{\text{coral}} + \lambda\mathcal{L}_{\text{cl}}$: Adding the unsupervised contrastive loss slightly improves performance on the BO dataset, but leads to declines on IP (76.93%) and SA (91.49%). This suggests that the lack of supervision may cause the false-negative problem, where similar samples are pushed apart and overall performance is degraded.
3) $\mathcal{L}_{\text{fsl}} + \mathcal{L}_{\text{coral}} + \lambda\mathcal{L}_{\text{scl}}$: Replacing the unsupervised loss with a supervised contrastive loss improves performance on the IP and SA datasets. However, AOA drops to 96.93% on the BO dataset, suggesting that while supervision alleviates the false-negative problem, class structure modeling remains suboptimal.
4) $\mathcal{L}_{\text{fsl}} + \mathcal{L}_{\text{coral}} + \lambda\mathcal{L}_{\text{mpc}}$*:* Introducing the MP contrastive loss leads to the best performance across all datasets. These results confirm that the momentum mechanism plays a critical role in constructing stable prototypes and reducing domain bias, thereby enhancing class structure modeling and cross-domain generalization.

2) *Module Ablation Experiments:* To validate the contribution of each key component to few-shot classification, a stepwise ablation study is conducted. Starting from the baseline model with only the FSL task, four modules are incrementally added: CORAL, prototype CL (PCL), AAM, and the MQ mechanism. As shown in Table X, five ablation variants are constructed as follows.
   1) *FSL:* The basic model performs only the few-shot classification task and serves as a reference.
   2) *FSL + CORAL:* CORAL loss is introduced to align covariance between domains. Despite achieving global feature alignment, performance drops on the IP and BO datasets due to the lack of class-level guidance, leading to class confusion and negative transfer.
   3) *FSL + CORAL + PCL:* Prototype CL is added to improve interclass separability. AOA increases to 78.46%, 97.57%, and 92.54% on the BO, SA, and IP datasets, respectively, confirming the effectiveness of PCL in enhancing feature discriminability.
   4) *FSL + CORAL + PCL + AAM:* The AAM introduces diverse contrastive views. Further gains are observed, with AOAs reaching 78.69%, 97.71%, and 92.57%, demonstrating improved representation robustness.
   5) *FSL + CORAL + PCL + AAM + MQ:* The MQ is incorporated to generate MP and reduce bias in cross-domain settings. The final model achieves AOAs of 80.28%, 97.92%, and 93.02% on BO, SA,

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TABLE VIII
AOA OF DIFFERENT HYPERPARAMETRIC MODEL

Dataset	m = 0.9	m = 0.99	m = 0.999	Cosine Schedule
IP	78.60 ± 3.00	79.63 ± 3.05	79.43 ± 3.15	**80.28 ± 3.26**
BO	97.52 ± 1.21	97.84 ± 1.27	97.80 ± 1.27	**97.92 ± 1.34**
SA	92.59 ± 1.25	92.75±0.97	92.91 ± 1.02	**93.02 ± 0.89**

TABLE IX
AOA OF ABLATION EXPERIMENTS ON LOSS

Loss Combination	IP	BO	SA
$\mathcal{L}_{fsl} + \mathcal{L}_{coral}$	77.03 ± 2.08	96.65 ± 1.43	92.20 ± 1.02
$\mathcal{L}_{fsl} + \mathcal{L}_{coral} + \lambda\mathcal{L}_{cl}$	76.93 ± 2.59	97.40 ± 1.21	91.49 ± 0.71
$\mathcal{L}_{fsl} + \mathcal{L}_{coral} + \lambda\mathcal{L}_{scl}$	80.05 ± 2.40	96.93 ± 1.38	92.69 ± 0.91
$\mathcal{L}_{fsl} + \mathcal{L}_{coral} + \lambda\mathcal{L}_{mpc}$	**80.28 ± 3.26**	**97.92 ± 1.34**	**93.02 ± 0.89**

and IP, respectively, indicating improved intraclass compactness, interclass separation, and semantic integrity.

I. Complexity Analysis

Table XI summarizes the training time (Tr), testing time (Te), and the number of parameters required during testing (Param) for all methods across the three datasets. Overall, DM-MRN shows the highest training and inference efficiency. Since its relation network models intradomain sample dependencies without requiring complex cross-domain feature alignment, faster execution is achieved. In contrast, most other methods involve cross-domain alignment or contrastive strategies, requiring computation of relationships between source and target features, which leads to longer training times. On the IP dataset, PGSMC shows slightly higher training time, mainly due to the introduction of the MQ mechanism, which increases RAM usage and introduces additional memory overhead. On the BO and SA datasets, longer training times are observed for HFSL and FSCF-SSL. This is primarily caused by the use of larger input patches, which result in memory consumption exceeding physical RAM limits. As a consequence, part of the data is offloaded to disk-based virtual memory, significantly reducing data transfer speed.

In terms of model size, DCFSL and MLPA adopt the same lightweight feature extractor, resulting in identical and relatively small parameter counts. DM-MRN, GCC-FSL, CTF-SSCL, and PGSMC employ custom architectures based on their respective designs, leading to moderate model sizes that balance performance and efficiency. In contrast, HFSL and FSCF-SSL incorporate more complex encoders and larger input patches, resulting in significantly larger model sizes due to increased architectural complexity.

V. DISCUSSION

This section will apply statistical significance tests to validate the practical contribution of MPC. Additionally, the performance and robustness of PGSMC will be evaluated in challenging scenarios. Finally, the distinct advantages of PGSMC in specific application contexts will be discussed.

A. Statistical Significance Testing

To validate the practical contribution of the MPC framework in PGSMC, we applied a paired t-test [61] to compare the performance improvement of our complete MPC model over the baseline model (FSL + CORAL) across ten independent experimental runs. Following standard statistical practices, the significance level was set to $\alpha = 0.05$. When the computed p-value is less than α, we reject the null hypothesis (H_0: no true performance difference between the two models) and consider the observed difference statistically significant. The paired t-test results in Table XII show that the p-values for the IP, BO, and SA datasets are 0.0088, 0.0277, and 0.0359, respectively, all of which are below 0.05, confirming the statistical significance of the performance improvement of the MPC framework over the FSL + CORAL baseline.

B. Performance and Robustness in Challenging Scenarios

Although the proposed PGSMC demonstrates outstanding performance across multiple cross-domain, few-shot classification tasks, its effectiveness is limited under certain extreme conditions. Specifically, HSIC generally faces key challenges in handling both intraclass variability and interclass similarity. The following analysis will examine these limiting factors from both perspectives to provide a deeper understanding of the model's underlying mechanisms.

1) Intraclass Variability: In real-world scenarios, a single land cover class may encompass multiple subclasses with significant spectral differences. For example, class 1 ("building") may include metal roofs, tile roofs, and asphalt roofs, each with distinct spectral responses. This high intraclass variability poses a challenge for methods that use a single prototype to represent the entire class. Class 15 in the IP dataset is a typical mixed class, as it contains land cover types with highly dispersed spectral characteristics, such as buildings, grass, trees, and driveways. PGSMC method computes an MP for each class. For class 15, the prototype ultimately converges to the "average" or "central" position within the feature space of all its subclasses (e.g., buildings, grass, and trees). As shown in Fig. 6, although the MP generated by PGSMC exhibits stronger representativeness compared to other CL methods, thus improving classification performance for class 15, these prototypes may not effectively represent specific subclasses. For instance, pure "building" pixels may be farther from the "average" prototype of class 15 and instead closer to the prototype of a distinct class (e.g., class 16, "steel tower"), leading to misclassification.

2) Interclass Similarity: Interclass spectral similarity is considered a significant challenge for classifiers that rely on spectral features. In the experiments conducted [Fig. 5(c)], both class 8 and class 15 in the SA dataset were utilized to represent vineyards. The characteristics of these classes include their geographical proximity and the similarity in the chemical and physical properties of their primary components (grapevines and soil), resulting in highly similar spectral features. As illustrated in Fig. 10, a failure to distinguish these classes is observed in instance-level CL methods (e.g., FSCF-SSL and CTF-SSCL). This is attributed to the tendency

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TABLE X
AOA OF ABLATION EXPERIMENTS ON THE MODULE

	Module					Dataset		
	FSL	CORAL	PCL	AAM	MQ	IP	BO	SA
1	✓					77.10 ± 2.32	96.91 ± 1.34	91.87 ± 1.08
2	✓	✓				77.03 ± 2.08	96.65 ± 1.43	92.20 ± 1.02
3	✓	✓	✓			78.46 ± 3.16	97.57 ± 1.29	92.54 ± 0.88
4	✓	✓	✓	✓		78.69 ± 3.33	97.71 ± 1.28	92.57 ± 0.95
5	✓	✓	✓	✓	✓	**80.28 ± 3.26**	**97.92 ± 1.34**	**93.02 ± 0.89**

TABLE XI
COMPLEXITY ANALYSIS FOR VARIOUS METHODS ON EACH DATASET

Dataset		DCFSL	HFSL	DM-MRN	FSCF-SSL	GCC-FSL	CTF-SSCL	MLPA	PGSMC
IP	Tr (s)	876.44	350.88	83.30	590.21	2152.58	983.20	935.42	1054.36
	Te (s)	0.53	2.45	12.81	2.52	0.47	1.04	0.49	0.48
	Param (K)	70.7	3304.4	211.7	3304.4	218.4	398.0	70.7	275.2
BO	Tr (s)	581.81	1626.27	116.59	2854.27	1472.79	384.52	654.81	793.66
	Te (s)	0.14	0.43	3.40	0.45	0.14	0.16	0.14	0.13
	Param (K)	64.9K	3221.2	211.7	3221.2	212.9	394.7	64.9	269.7
SA	Tr (s)	895.13	1535.95	96.20	2677.33	2184.37	747.95	956.05	1083.49
	Te (s)	2.67	155.57	65.71	145.33	2.60	2.68	2.61	2.60
	Param (K)	71.1	3310.6	211.7	3310.6	218.8	400.7	71.1	275.6

TABLE XII
PAIRED T-TEST RESULTS FOR THE HOLISTIC CONTRIBUTION OF THE MPC FRAMEWORK

Datasets	IP	BO	SA
p-value	0.0088	0.0277	0.0359

of these methods to reduce the distance between spectrally similar pairs, thereby exacerbating confusion. In contrast, better performance is demonstrated in the t-SNE visualization of PGSMC, where more compact clusters and reduced overlap are shown. However, a complete resolution of this issue is not achieved by PGSMC. When inherent separability is lacking in the features extracted by the EM from the raw hyperspectral data, the computed MPs become positioned too closely in the feature space. Although repulsive forces are imposed by the loss function to separate them, its effectiveness is constrained by the inherent separability of the features. If sufficient gradient information is not provided, a partial merging of these two classes by the model may still occur.

C. Advantages in Specific Applications

PGSMC demonstrates significant advantages in cross-domain few-shot classification tasks, particularly in knowledge transfer across sensors and domains, and class preservation. Unlike traditional same-domain FSL methods, such as S3Net and DM-MRN, which assume similar distributions between source and target domains, PGSMC effectively bridges the distribution gap through PCL and MPs, making it particularly suitable for fields like remote sensing image classification. Compared to other DA methods, such as DCFSL and MLPA, PGSMC improves interclass separation and intraclass compactness, enhancing classification performance in tasks that require cross-DA and consistent class structure, especially when labeled samples in the target domain are scarce. Moreover, PGSMC's use of category-level CL with MPs, as opposed to traditional instance-level methods like GCC-FSL and CTF-SSCL, effectively maintains class boundaries and reduces confusion, making it ideal for multiclass classification tasks, such as land cover classification.

VI. CONCLUSION

In this article, a novel MPC learning framework is proposed for HSIC to improve performance under few-shot conditions. First, AAM is introduced to enhance feature diversity and robustness through asymmetric augmentation strategies and a momentum-updated encoder. Then, MPCM is incorporated to provide long-term memory by smoothly updating feature representations of each sample. Finally, $\mathcal{L}_{\text{mpc}}$ is designed to encourage the model to focus directly on more representative class prototypes, thereby improving interclass separability. Extensive experiments confirm the effectiveness of the proposed PGSMC in addressing HSIC tasks. In future work, efforts will be devoted to further optimizing the framework and advancing the algorithm design to explore more efficient FSL architectures.

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Lingyu Kong received the B.S. degree in software engineering from Hunan University of Humanities, Science and Technology, Loudi, China, in 2023. He is currently pursuing the M.S. degree in software engineering with the College of Computer Science and Technology, Qingdao University, Qingdao, China.

His research interests include deep learning and hyperspectral few-shot learning.



Xudong Sun (Member, IEEE) received the M.S. and Ph.D. degrees in computer science and technology from Dalian Maritime University, Dalian, China, in 2020 and 2024, respectively.

He is currently working with the College of Computer Science and Technology, Qingdao University, Qingdao, China. His research interests include hyperspectral image processing, imaging science and photographic technology, and underwater robotics.



Zifei Zhao received the B.S. degree in software engineering from Hunan University of Humanities, Science and Technology, Loudi, China, in 2024. He is currently pursuing the M.S. degree in software engineering with the College of Computer Science and Technology, Qingdao University, Qingdao, China.

His research interests include artificial intelligence and hyperspectral image classification.



Jiahua Zhang received the Ph.D. degree in cartography and remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, China, in 1998.

From 1999 to 2001, he held a postdoctoral position with the National Institute for Environmental Studies, Tsukuba, Japan. Since January 2002, he has been a Professor with Chinese Academy of Meteorological Sciences, Beijing. Since 2012, he has been a Full Professor with the Institute of Remote Sensing and Digital Earth, CAS, and now a Professor with Qingdao University, Qingdao, China. He has published more than 200 peer-reviewed articles, 30 international conferences papers, and six books. His research interests include remote sensing and geosciences, vegetation dynamics, land use and land cover classification, image processing, deep learning, and coastal environment remote sensing.



Xiaodi Shang (Member, IEEE) received the Ph.D. degree in computer application technology from Dalian Maritime University, Dalian, China, in 2022.

She is currently an Associate Professor with the College of Computer Science and Technology, Qingdao University, Qingdao, China. Her research interests include machine learning and hyperspectral image processing.

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